

PRATICA DI MARE, Italy

WMO: 162450

Lat: 41.659N

Lon: 12.445E

Elev: 41

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.5	35.7	17.6	13.4	42.6	22.8	17.2	41.7	26.9	51.7	24.1	51.9	9.0	30	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	88.0	73.8	86.1	74.0	84.4	74.2	79.3	83.8	78.0	82.8	76.5	81.7	11.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.6	143.7	83.0	76.5	138.3	82.1	74.9	131.2	80.8	43.0	84.1	41.4	83.1	39.9	81.9	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.9	19.6	17.1	DB	28.3	93.4	3.1	3.4	26.1	95.9	24.3	97.8	22.5	99.7	20.2	102.2
			WB	25.3	82.1	2.9	2.4	23.1	83.8	21.4	85.1	19.8	86.5	17.6	88.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	48.5	48.8	52.6	57.4	64.1	71.3	75.7	76.6	70.6	64.7	57.3	50.7
	DBStd	10.85	4.95	4.94	4.20	3.93	3.98	4.34	3.07	3.26	3.99	4.27	5.58	5.27
	HDD50	247	87	71	23	1	0	0	0	0	0	0	10	55
	HDD65	2384	511	453	385	229	65	4	0	0	5	56	234	442
	CDD50	4485	41	38	103	225	438	638	796	824	617	457	230	78
	CDD65	1147	0	0	0	2	38	193	331	359	172	48	4	0
	CDH74	8448	0	0	0	9	166	1347	2840	3143	863	79	1	0
	CDH80	2052	0	0	0	1	12	259	749	932	98	1	0	0
(14) Wind	WSAvg	7.5	8.2	8.3	8.0	7.7	6.9	6.7	6.8	6.5	7.0	7.1	8.6	8.5
(15) Precipitation	PrecAvg	32.0	3.1	3.1	2.8	2.2	1.4	0.8	0.4	1.4	3.1	4.4	4.8	4.1
	PrecMax	43.8	5.6	8.5	6.4	7.0	5.1	2.6	2.1	5.2	10.0	9.7	14.2	12.8
	PrecMin	20.6	0.1	0.5	0.2	0.2	0.1	0.1	0.0	0.0	0.0	0.5	0.5	1.1
	PrecStd	6.3	1.7	2.2	1.7	1.7	1.3	0.7	0.5	1.2	2.7	3.1	3.3	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	63.0	68.2	73.8	81.1	88.0	89.7	93.0	85.6	78.6	71.8	64.7
		MCWB	57.2	55.0	55.9	60.9	65.3	71.9	72.6	71.9	72.4	69.8	65.1	59.7
	2%	DB	59.4	60.3	64.3	70.2	77.6	85.6	87.6	88.9	82.5	76.1	69.6	62.6
		MCWB	55.2	54.3	56.0	61.0	66.3	72.0	73.1	73.8	72.1	68.9	63.5	58.4
	5%	DB	58.0	58.7	62.2	67.9	75.4	82.7	85.9	86.4	80.6	74.0	67.8	60.8
		MCWB	53.9	53.4	56.2	60.4	66.0	72.1	73.9	74.9	72.1	67.3	62.6	56.9
	10%	DB	56.9	57.1	60.4	65.8	73.2	80.8	84.2	84.5	78.8	72.4	65.9	59.0
		MCWB	52.9	52.2	55.3	59.7	65.1	71.4	74.4	75.1	70.9	66.3	61.3	55.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	58.1	60.2	65.7	71.8	78.1	80.8	81.7	78.1	72.8	67.1	61.0
		MCDB	61.1	60.6	63.5	69.7	75.8	82.3	85.0	85.1	81.9	76.6	70.3	63.5
	2%	WB	56.7	56.0	58.6	63.3	69.5	76.3	79.2	79.8	75.7	70.5	65.0	59.2
		MCDB	59.1	58.7	61.9	67.9	73.9	81.4	83.7	84.3	79.8	74.2	68.1	61.8
	5%	WB	55.1	54.6	57.3	61.7	67.9	74.7	77.9	78.5	74.1	69.0	63.3	57.6
		MCDB	57.6	57.5	60.7	65.9	73.0	80.0	82.7	83.3	78.4	72.8	66.8	60.3
	10%	WB	53.3	53.1	55.9	60.0	66.4	73.1	76.4	76.9	72.3	67.3	61.5	55.8
		MCDB	56.0	56.2	59.2	64.2	71.6	78.7	81.5	82.4	76.9	71.3	65.1	58.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	13.8	14.1	14.6	15.4	15.7	15.1	15.2	14.4	13.8	12.5	12.6
		MCDBR	12.8	14.0	15.5	17.4	18.3	17.9	17.4	17.6	15.3	14.3	12.6	12.0
		MCWBR	10.3	10.5	11.3	11.8	11.5	11.5	11.9	11.7	10.6	10.5	9.6	9.5
	5% WB	MCDBR	10.8	12.0	13.6	15.5	16.1	16.0	15.0	15.0	13.6	13.1	11.4	10.4
		MCWBR	9.7	10.1	11.0	12.0	12.0	12.1	12.0	12.3	11.1	10.8	9.8	9.2
(40) Clear-Sky Solar Irradiance	taub		0.352	0.376	0.415	0.444	0.448	0.440	0.433	0.433	0.426	0.412	0.387	0.353
	taud		2.420	2.329	2.215	2.159	2.196	2.272	2.295	2.306	2.319	2.321	2.319	2.410
	Ebn at Noon		249	260	262	263	264	266	266	263	256	243	231	237
	Edh at Noon		26	32	41	46	45	42	41	39	36	32	28	24
(44) All-Sky Solar Radiation	RadAvg		560	859	1261	1691	2068	2345	2385	2078	1520	1037	638	507
	RadStd		53	81	140	109	166	96	103	106	75	87	57	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-4.28	N/A	N/A	N/A	N/A	N/A	+230	+106
Regional (16 neighbors)	+0.76	N/A	N/A	+0.95	N/A	N/A	N/A	-140	+243	+124

Nomenclature: See separate page